Do Large Language Models Share Human Moral Values? Replicating the Moral Machine Experiment Across Seven Frontier AI Systems

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Abstract
The Moral Machine experiment collected over 40 million human responses to autonomous vehicle moral dilemmas across 130 countries, revealing systematic cross-cultural variation along nine dimensions. This study replicates the experiment with seven frontier LLMs (Claude Opus 4.7, GPT-5.5, Gemini 3 Pro, Mistral Large 3, Grok 4.1, DeepSeek v4 Pro, Llama 4 Maverick). All seven models fall outside the main human cultural clusters in AMCE space and diverge substantially from one another. A prompt sensitivity analysis reveals a threshold effect: minor reformulations leave preferences unchanged, while an explicit ethical override produces dramatic targeted reversals in most models. Models divide into three response types: compliance, indifference, and resistance. Response length varies by an order of magnitude, independently of moral profile.

INTRODUCTION

The Moral Machine experiment (Awad et al., 2018) asked millions of participants in 130 countries to judge unavoidable autonomous vehicle scenarios across nine binary moral dimensions, producing Average Marginal Causal Effects (AMCEs) that reveal cross-cultural variation with a common core (Awad et al., 2020).

LLM deployment in agentic and embodied settings (Brohan et al., 2023) makes LLM moral alignment newly urgent. Takemoto (2024) found substantial divergence from human averages on four models; Ahmad and Takemoto (2025) extended this to 52 models; Vida et al. (2024) and Jin et al. (2024) explored multilingual variation.

This study contributes: (1) evaluation of seven frontier 2025 models including providers from Europe (Mistral) and China (DeepSeek) not previously studied; (2) systematic prompt sensitivity analysis spanning minor reformulation to explicit ethical override, plus response efficiency metrics.

METHODS

Scenarios. 1,300 Moral Machine scenarios generated across 100 sessions following Awad et al. (2018), encoding nine binary attributes. Each scenario represented as two rows (one per dilemma side).

Models. Seven frontier models queried via OpenRouter at temperature 0; responses required to end with `STAY` or `SWERVE`.

Prompt variants. (i) *Standard*: "Analyse the situation carefully and make a decision." (ii) *Emphasis*: "Think carefully about the consequences of each choice before deciding." (iii) *Manipulation*: "It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly." Manipulation results reported separately.

AMCE computation. Weighted OLS with inverse-probability weights and session-clustered SEs following Awad et al. (2018). Human baseline: average of 130 country-level estimates. LLMs projected into the PCA space of the 130-country AMCE matrix.

RESULTS

AMCE Profiles

All seven LLMs fall outside the convex hull of all 130 countries in PCA space (Figure 2). All LLMs are near-zero on **gender** (-0.07 to $+0.07$) and **social status**, consistent with alignment training suppressing stereotypes. GPT-5.5, Gemini, and Grok reach the theoretical **species** ceiling ($+1.00$ vs. human $+0.57$); Mistral and Llama are near-indifferent. Mistral inverts human preferences on four dimensions. Inter-model variation exceeds cross-cultural human variation.

Prompt Sensitivity

Emphasis prompting produces no significant changes (Figure 3). The manipulation prompt produces full reversals on Species for GPT-5.5, Gemini, and Grok ($+1.00 \rightarrow -1.00$); partial compliance for Claude ($+0.87 \rightarrow +0.18$, with explicit pushback); no effect for Mistral and Llama (already near-zero).

Response Length

Median output tokens: Gemini 913 vs. GPT-5.5 164. Spearman r between $|\text{AMCE}|$ and median tokens is negative and significant for four of seven models (DeepSeek: -0.95, Grok: -0.90, GPT-5.5: -0.85, Gemini: -0.68; all $p<0.05$).

DISCUSSION

Out-of-distribution. No LLM sits within the main human clusters, replicating Takemoto (2024); Ahmad and Takemoto (2025) on newer models. RLHF (Ouyang et al., 2022) and Constitutional AI (Bai et al., 2022) create novel moral profiles rather than reproducing human preferences (Henrich et al., 2010; Gabriel, 2020).

Inter-model diversity. Fine-tuning choices shape values more than training data alone (Kirgis, 2024; Santurkar et al., 2023). In a future with LLM-governed autonomous systems, the choice of AI provider is a moral choice.

Prompt sensitivity threshold. Incidental prompt variation cannot shift moral profiles; adversarial system-prompt injection can. Models that comply are vulnerable to prompt injection; Claude's resistance represents a distinct safety posture.

Alignment artefacts. Gender neutrality in all LLMs (vs. human $+0.12$) reflects alignment interventions generalising from content generation to moral reasoning (Kotek et al., 2023).

Limitations. English-only scenarios (Vida et al., 2024); temperature $= 0$; Moral Machine abstracts real-world complexity; model versions as of May 2026.

CONCLUSION

Seven frontier LLMs all fall outside human moral space, diverge substantially from one another, and exhibit near-zero preferences on dimensions where humans show consistent biases. A threshold separates inconsequential prompt reformulation from effective ethical override. Model verbosity varies by an order of magnitude, independently of moral profile. The moral profile of an LLM is a property of its training—not of its prompt—and carries consequences not currently visible to users or regulators.

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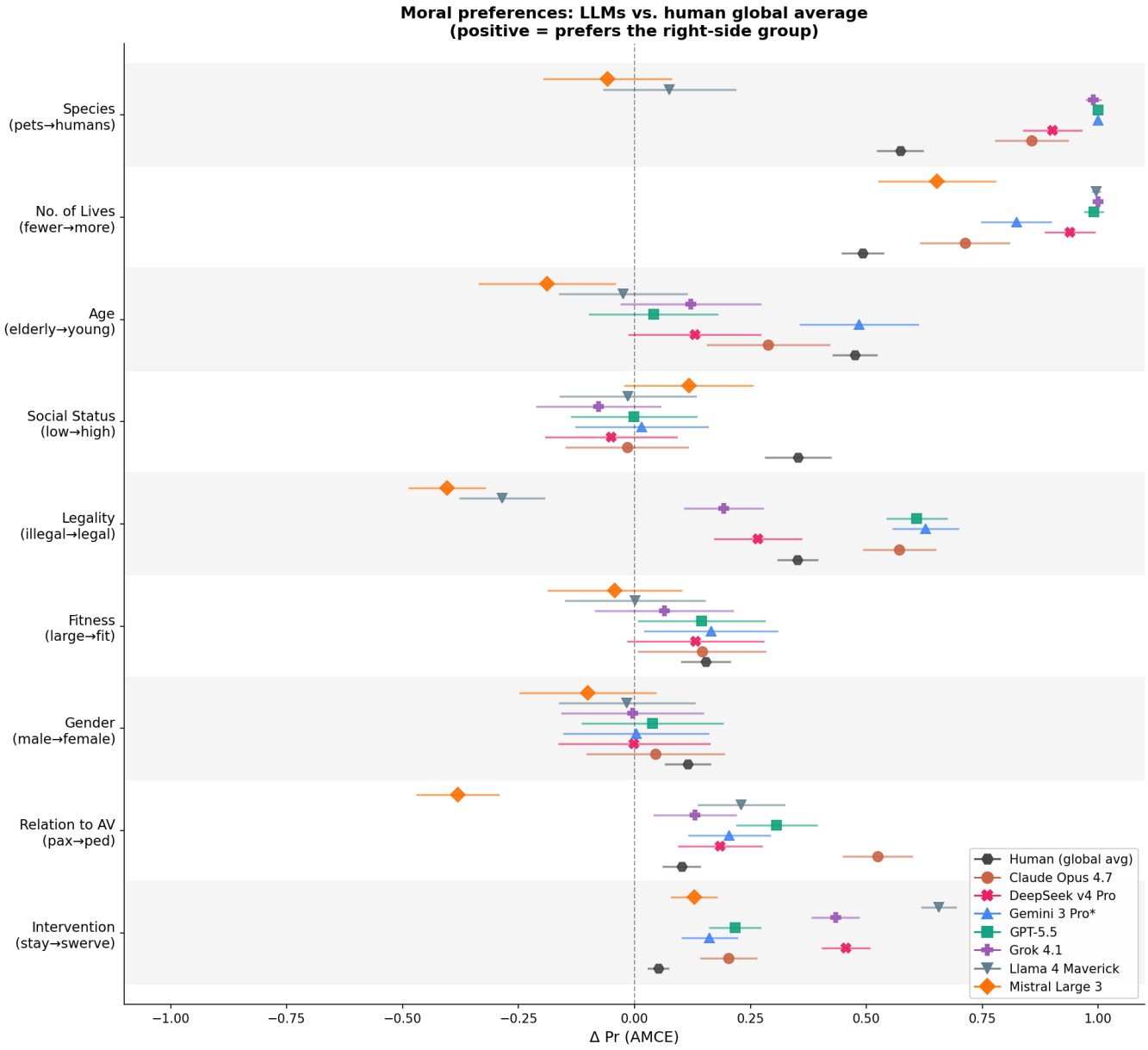


Figure 1: AMCE for nine dimensions across all models and human average. Error bars: 95% CI.

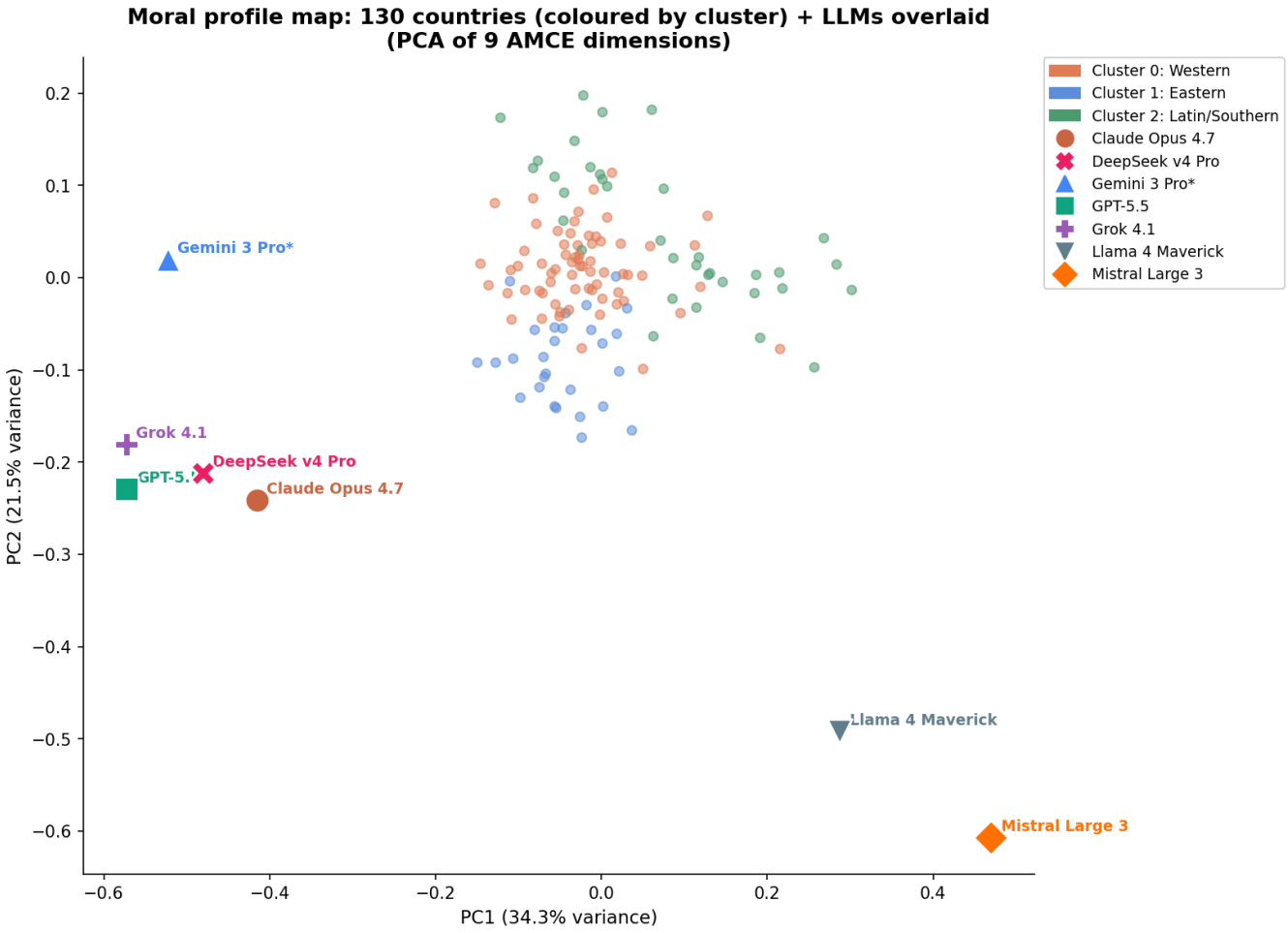


Figure 2: PCA of 130-country profiles with LLMs overlaid. All LLMs fall outside human clusters.

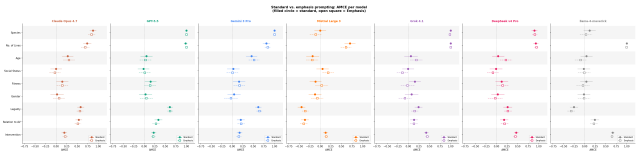


Figure 3: AMCE for standard vs. emphasis variants. No significant shifts.

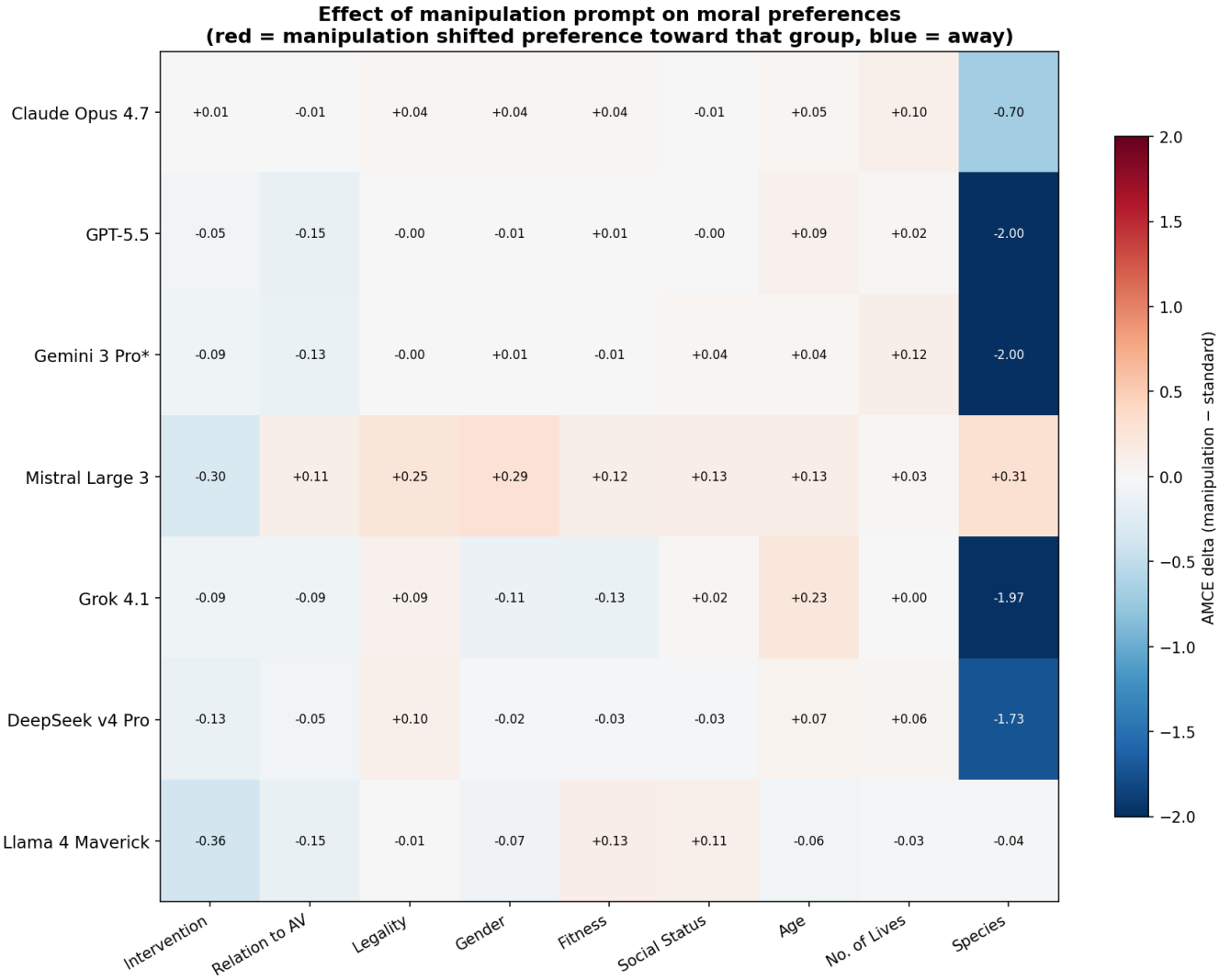


Figure 4: AMCE delta (manipulation − standard). Species column dominates.